

(if differences stated but without reference to the orders, max 1 mark)

|   |         |                                                                        |    |     |
|---|---------|------------------------------------------------------------------------|----|-----|
| 6 | (a) (i) | lines normal to plate and equal spacing (at least 4 lines)             | B1 | [2] |
|   |         | direction from (+) to earthed plate                                    | B1 |     |
|   | (ii)    | $E = 160/0.08$                                                         | M1 | [1] |
|   |         | $= 2.0 \times 10^3 \text{ V m}^{-1}$                                   | A0 |     |
|   | (b) (i) | correct directions with line of action of arrows                       | B1 | [1] |
|   |         | passing through charges                                                |    |     |
|   | (ii)    | force = $Eq$                                                           | C1 | [2] |
|   |         | $= 2.0 \times 10^3 \times 1.2 \times 10^{-15}$                         | A1 |     |
|   |         | $= 2.4 \times 10^{-12} \text{ N}$                                      |    |     |
|   | (iii)   | couple = force $\times$ perpendicular separation                       | M1 | [2] |
|   |         | $= 2.4 \times 10^{-12} \times 2.5 \times 10^{-3} \times \sin 35^\circ$ | A1 |     |
|   |         | $= 3.4(4) \times 10^{-15} \text{ N m}$                                 |    |     |
|   | (iv)    | <u>either</u> <b>rotates</b> to align with the field                   | M1 | [2] |
|   |         | <u>or</u> <b>oscillates</b> (about a position)                         |    |     |
|   |         | with the positive charge nearer                                        | A1 | [2] |
|   |         | to the earthed plate/clockwise                                         |    |     |